TED (15) - 3021 Reg. No.........................................

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**MODEL QUESTION PAPER**

# THIRD SEMESTER DIPLOMA EXAMINAHON IN ENGINEERING/ TECHNOLOGY - OCTOBER, 2016

**ELECTRICAL AND ELECTRONICS ENGINEERING**

(Common for MH, AU and TD)

*[Time:* 3 hours

**(**Maximum marks : 100)

PART—A

(Maximum marks : 10)

Marks

I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Define form factor
2. Define Ampere hour efficiency of a lead acid cell.
3. Give any two application of induction motor
4. List two types of moving iron instrument
5. Draw the symbol of NOR gate and develop the truth table.

(5X2=10)

PA RT—B

(Maximum marks : 30)

II Answer any five questions. Each question carries 6 marks.

1. Show the effective resistance when three resistors are connected in
   (a) series (b) parallel
2. Summarizes the advantages of a 3-phase system over single phase system
3. Draw 3 point Starter and mark necessary parts
4. Differentiate the welding transformer and power transformer
5. With necessary figure, explain the constructional details of Moving Iron instruments
6. Distinguish different types of capacitors
7. Summarize the need of automation

(5X6=30)

PART—C

(Maximum marks : 60)

(Answer one full question from each module. Each question carries 15 marks.)

Module - I

III a) A resistor of 3 Ω is connected in parallel with a resistor of 6 Ω. In series with the parallel combination is a 8Ω resistor and applied a 20V to the circuit. Compute the effective resistance, current and power in the circuit 8

b) Describe the methods of charging of lead acid cell. 7

OR

IV a) Explain the classification of D.C. generators based on field connection 8

b) Describe the power generation of three phase system. 7

Module - II

V a) Explain the working principle of single phase capacitor run induction motor 8

b) Determine the e.m.f equation of transformer. 7

OR

VI a) Explain the constructional details of single phase induction motor 8

b) Describe the principle of working of D C Motors 7

Module - III

V a) Explain working principle of dynamometer types wattmeter 8

b) Explain the principle of induction heating. 7

OR

VI a) Explain the constructional details of Moving Coil instruments. 8

b) Explain the principle of induction heating. 7

Module - IV

V a) Draw and explain working of full wave bridge rectifier 8

b) List the applications of control system 7

OR

VI a) Explain the working principle of SCR. 8

b) Describe the working of PN junction diode. 7